8. In the proof of Theorem D we used the following fact: the image of a Cauchy sequence under a uniformly continuous mapping is again a Cauchy sequence. Give the details of the proof.
9. Let f be a continuous real function defined on R which satisfies the functional equation $f(x + y) = f(x) + f(y)$. Show that this function must have the form $f(x) = mx$ for some real number m. (*Hint:* the subspace of rational numbers is dense in the metric space R.)

14. SPACES OF CONTINUOUS FUNCTIONS

In Example 9-5 we gave a brief description of the metric space $\mathcal{C}[0,1]$. The reader will recall that the points of this space are the bounded continuous real functions defined on the closed unit interval $[0,1]$ and that its metric is defined by $d(f,g) = \sup |f(x) - g(x)|$. We have two aims in this section: to generalize this very important example by considering functions defined on an arbitrary metric space, and to place all function spaces of this type in their proper context by giving the details of the structural pattern (discussed briefly in Sec. 9) which they all have in common with one another. We begin with the second, and define the algebraic systems which are relevant to our present interests.

Let L be a non-empty set, and assume that each pair of elements x and y in L can be combined by a process called *addition* to yield an element z in L denoted by $z = x + y$. Assume also that this operation of addition satisfies the following conditions:

(1) $x + y = y + x$;
(2) $x + (y + z) = (x + y) + z$;
(3) there exists in L a unique element, denoted by 0 and called the *zero element*, or the *origin*, such that $x + 0 = x$ for every x;
(4) to each element x in L there corresponds a unique element in L, denoted by $-x$ and called the *negative* of x, such that $x + (-x) = 0$.

We adopt the device of referring to the system of real numbers or to the system of complex numbers as the *scalars*. We now assume that each scalar α and each element x in L can be combined by a process called *scalar multiplication* to yield an element y in L denoted by $y = \alpha x$ in such a way that

(5) $\alpha(x + y) = \alpha x + \alpha y$;
(6) $(\alpha + \beta)x = \alpha x + \beta x$;
(7) $(\alpha\beta)x = \alpha(\beta x)$;
(8) $1 \cdot x = x$.

The algebraic system L defined by these operations and axioms is called a *linear space.* Depending on the numbers admitted as scalars (only the real numbers, or all the complex numbers), we distinguish when necessary between *real linear spaces* and *complex linear spaces.* For geometric reasons discussed in the next section, a linear space is often called a *vector space,* and its elements are spoken of as *vectors.*

We are not concerned here with developing the algebraic theory of linear spaces. Our only interest is in making available some pertinent concepts and terminology which are useful as a background against which to view the metric spaces we wish to study. With this in mind, we mention a few simple facts which are quite easy to prove from the axioms for a linear space: $0 + x = x$ for every x; $x + z = y + z \Rightarrow x = y$ (*hint:* add $-z$ to both sides on the right); $\alpha \cdot 0 = 0$ (*hint:* $\alpha \cdot 0 + \alpha x = \alpha(0 + x) = \alpha x = 0 + \alpha x$); $0 \cdot x = 0$ (*hint:* $0 \cdot x + \alpha x = (0 + \alpha)x = \alpha x = 0 + \alpha x$); and $(-1)x = -x$ (*hint:* $x + (-1)x = 1 \cdot x + (-1)x = (1 + (-1))x = 0 \cdot x = 0$). The reader will notice that in the relation $0 \cdot x = 0$ we have used the symbol 0 with two different meanings: as a scalar on the left and as a vector on the right. Several other meanings will be given to this single symbol, but fortunately it is always possible to avoid confusion by attending closely to the context in which it occurs. It is convenient to introduce the operation of *subtraction* by using the symbol $x - y$ as an abbreviation for $x + (-y)$; $x - y$ is called the *difference* between x and y.

A non-empty subset M of a linear space L is called a *linear subspace* of L if $x + y$ is in M whenever x and y are and if αx is in M (for any scalar α) whenever x is. Since M is non-empty, $0 \cdot x = 0$ shows that 0 is in M. Since $-x = (-1)x$, $-x$ is in M whenever x is. It will be seen at once that a linear subspace of a linear space is itself a linear space with respect to the same operations.

A *normed linear space* is a linear space on which there is defined a *norm,* i.e., a function which assigns to each element x in the space a real number $\|x\|$ in such a manner that

(1) $\|x\| \geq 0$, and $\|x\| = 0 \Leftrightarrow x = 0$;
(2) $\|x + y\| \leq \|x\| + \|y\|$;
(3) $\|\alpha x\| = |\alpha| \, \|x\|$.

In general terms, a normed linear space is simply a linear space in which there is available a satisfactory notion of the distance from an arbitrary element to the origin. From (3) and the fact that $-x = (-1)x$, we obtain $\|-x\| = \|x\|$. As we saw in Sec. 9, a normed linear space is a metric space with respect to the induced metric defined by

$$d(x,y) = \|x - y\|.$$

A *Banach space* is a normed linear space which is complete as a metric

space. By Theorem 12-B, any closed linear subspace of a Banach space is itself a Banach space with respect to the same algebraic operations and the same norm.

So much for the technical framework. We now turn to the metric spaces which really concern us. They are all *function spaces*, in the sense that they are linear spaces whose elements are functions defined on some non-empty set X with addition and scalar multiplication defined pointwise, i.e., by $(f + g)(x) = f(x) + g(x)$ and $(\alpha f)(x) = \alpha f(x)$. We note that the zero element in such a linear space is the constant function 0 whose only value is the scalar 0 and that $(-f)(x) = -f(x)$.

Suppose, then, that X is an arbitrary non-empty set, and consider the set L of all real functions defined on X. It is clear that L is a real linear space with respect to the operations described above. We now restrict ourselves to the subset B consisting of the bounded functions in L. B is obviously a linear subspace of L, so it is a linear space in its own right. Even more, if we define a norm on B by $\|f\| = \sup |f(x)|$, then B is a Banach space (see Problems 9-5 and 12-4).

We next assume that the underlying set X is a metric space. This enables us to consider the possible continuity of functions defined on X. We define $\mathcal{C}(X,R)$ to be that subset of B which consists of continuous functions. $\mathcal{C}(X,R)$ is thus the set of all bounded continuous real functions defined on the metric space X, and it is non-empty by Problem 13-1.

Lemma. *If f and g are continuous real functions defined on a metric space X, then $f + g$ and αf are also continuous, where α is any real number.*

PROOF. Let d be the metric on X. We show that $f + g$ is continuous by showing that it is continuous at an arbitrary point x_0 in X. Let $\epsilon > 0$ be given. Since f and g are continuous, and thus continuous at x_0, we can find $\delta_1 > 0$ and $\delta_2 > 0$ such that $d(x,x_0) < \delta_1 \Rightarrow |f(x) - f(x_0)| < \epsilon/2$ and $d(x,x_0) < \delta_2 \Rightarrow |g(x) - g(x_0)| < \epsilon/2$. Let δ be the smaller of the numbers δ_1 and δ_2. Then the continuity of $f + g$ at x_0 follows from

$$d(x,x_0) < \delta \Rightarrow |(f + g)(x) - (f + g)(x_0)| = |[f(x) + g(x)] - [f(x_0) + g(x_0)]| = |[f(x) - f(x_0)] + [g(x) - g(x_0)]| \leq |f(x) - f(x_0)| + |g(x) - g(x_0)| < \epsilon/2 + \epsilon/2 = \epsilon.$$

We leave it to the reader to show similarly that αf is continuous.

This lemma implies that $\mathcal{C}(X,R)$ is a linear subspace of the linear space B. We next prove that it is closed as a subset of the metric space B.

Lemma. *$\mathcal{C}(X,R)$ is a closed subset of the metric space B.*

PROOF. Let f be a function in B which is in the closure of $\mathcal{C}(X,R)$. We show that f is continuous, and therefore is in $\mathcal{C}(X,R)$, by showing that it is continuous at an arbitrary point x_0 in X. Since a set which equals its closure is closed, this will suffice to prove the lemma. Let d be the metric on X, and let $\epsilon > 0$ be given. Since f is in the closure of $\mathcal{C}(X,R)$, there exists a function f_0 in $\mathcal{C}(X,R)$ such that $\|f - f_0\| < \epsilon/3$, from which it follows that $|f(x) - f_0(x)| < \epsilon/3$ for every point x in X. Since f_0 is continuous, and hence continuous at x_0, we can find a $\delta > 0$ such that $d(x,x_0) < \delta \Rightarrow |f_0(x) - f_0(x_0)| < \epsilon/3$. The fact that f is continuous at x_0 now follows from

$$\begin{aligned} d(x,x_0) < \delta \Rightarrow |f(x) - f(x_0)| &= |[f(x) - f_0(x)] + [f_0(x) - f_0(x_0)] \\ &\quad + [f_0(x_0) - f(x_0)]| \leq |f(x) - f_0(x)| + |f_0(x) - f_0(x_0)| \\ &\quad + |f_0(x_0) - f(x_0)| < \epsilon/3 + \epsilon/3 + \epsilon/3 = \epsilon. \end{aligned}$$

Since a closed linear subspace of a Banach space is itself a Banach space, we can summarize the result of the above discussion and lemmas in the following theorem.

Theorem A. *The set $\mathcal{C}(X,R)$ of all bounded continuous real functions defined on a metric space X is a real Banach space with respect to pointwise addition and scalar multiplication and the norm defined by $\|f\| = \sup |f(x)|$.*

It is desirable at this stage to make a clear distinction between two types of convergence for sequences of functions. Let X be a metric space, and let $\{f_n\}$ be a sequence of real functions defined on X. If for each x in X it happens that $\{f_n(x)\}$ is a Cauchy sequence of real numbers, then by the completeness of the real number system we can define a limit function f by $f(x) = \lim f_n(x)$. We then say that f_n *converges pointwise* to f, or that f is the *pointwise limit* of f_n. It is often important to know what properties of the functions f_n carry over to the limit function f, but unless we strengthen the mode of convergence, very little can be said along these lines. The stronger type of convergence normally needed to conclude anything of interest is called *uniform convergence*. In order to explain what this is, we inspect a little more closely what is involved in pointwise convergence. To say that f_n converges pointwise to f is to say the following: for each point x in X, if $\epsilon > 0$ is given, then a positive integer n_0 can be found such that $|f_n(x) - f(x)| < \epsilon$ for all $n \geq n_0$. In general, the integer n_0 may depend on x as well as ϵ. If, however, for each given ϵ an integer n_0 can be found which serves for *all* points x, then we say that f_n *converges uniformly* to f, or that f is the *uniform limit* of f_n. The reader will observe that these concepts are quite independent of the assumption that X is a metric space and that they are meaningful for functions defined on an arbitrary non-empty set.

It will be seen at once that convergence in the function space $\mathcal{C}(X,R)$ is precisely uniform convergence as we have just defined it. The fact that $\mathcal{C}(X,R)$ is complete can be restated as follows in the language of uniform convergence: if a bounded real function f defined on X is the uniform limit of a sequence $\{f_n\}$ of bounded *continuous* real functions defined on X, then f is also continuous. In other words, in the presence of uniform convergence, continuity carries over from the f_n's to the limit function f.

A moment's thought will convince the reader that the entire discussion given above, beginning with our definition of the linear space L, could perfectly well have been based on complex functions. Without going again through all the details, we state the following theorem and consider it proved.

Theorem B. *The set $\mathcal{C}(X,C)$ of all bounded continuous complex functions defined on a metric space X is a complex Banach space with respect to pointwise addition and scalar multiplication and the norm defined by*

$$\|f\| = \sup |f(x)|.$$

In summary, we associate with each metric space X two linear spaces of continuous functions defined on X. The first—$\mathcal{C}(X,R)$—contains only real functions, and the second—$\mathcal{C}(X,C)$—consists of complex functions. Further, all functions considered are assumed to be bounded, so that the norm defined by $\|f\| = \sup |f(x)|$ is always a real number. In the special case in which X is a closed interval $[a,b]$ on the real line, we write $\mathcal{C}(X,R)$ in the simpler form $\mathcal{C}[a,b]$.

Problems

1. Show that a non-empty subset A of a Banach space is bounded $\Leftrightarrow$ there exists a real number K such that $\|x\| \leq K$ for every x in A.
2. Construct a sequence of continuous functions defined on [0,1] which converges pointwise but *not* uniformly to a continuous limit.
3. Construct a sequence of continuous functions defined on [0,1] which converges pointwise to a discontinuous limit.
4. Let X and Y be metric spaces with metrics d_1 and d_2, and let $\{f_n\}$ be a sequence of mappings of X into Y which converges pointwise to a mapping f of X into Y, in the sense that $f_n(x) \to f(x)$ for each x in X. Define what ought to be meant by the statement that f_n converges uniformly to f, and prove that under this assumption f is continuous if each f_n is continuous.
5. In this problem we give a procedure for constructing the *completion* X^* of an arbitrary metric space X. Denote by d the metric on X.

Let x_0 be a fixed point in X, and to each point x in X make correspond the real function f_x defined on X by $f_x(y) = d(y,x) - d(y,x_0)$.

(*a*) Show that f_x is bounded. (*Hint:* $|f_x(y)| \leq d(x,x_0)$.)

(*b*) Show that f_x is continuous. (*Hint:* $|f_x(y_1) - f_x(y_2)| \leq 2d(y_1,y_2)$.)

By (*a*) and (*b*), the mapping F defined by $F(x) = f_x$ is a mapping of X into $\mathfrak{C}(X,R)$.

(*c*) Show that F is an isometry. (*Hint:* $|f_{x_1}(y) - f_{x_2}(y)| \leq d(x_1,x_2)$.)

F is thus an isometry of X into the complete metric space $\mathfrak{C}(X,R)$. We define the *completion* X^* of X to be the closure of $F(X)$ in $\mathfrak{C}(X,R)$.

(*d*) Show that X^* is a complete metric space which contains an isometric image of X.

(*e*) Show that there is a natural isometry of X^* into any complete metric space Y which contains an isometric image of X (to say that an isometry of X^* into Y is "natural" means that the image of a point in X^* which corresponds to a point in X is the point in Y which corresponds to this same point in X).

(*f*) Show that (*d*) and (*e*) characterize X^* in the following sense: if Z is a complete metric space which contains an isometric image of X, and if there is a natural isometry of Z into any complete metric space Y which contains an isometric image of X, then there is a natural isometry of Z onto X^*.

(*g*) Show that if X occurs as a subspace of a complete metric space, then there is a natural isometry of the closure of X onto X^*.

(*h*) Show that there is a natural isometry of any complete metric space which contains X as a dense subspace onto X^*.[1]

15. EUCLIDEAN AND UNITARY SPACES

Let n be a fixed positive integer, and consider the set R^n of all ordered n-tuples $x = (x_1, x_2, \ldots, x_n)$ of real numbers.[2] We promised in Sec. 4 to make this set into a space, and we are now in a position to do so.

[1] The construction outlined in (*a*) to (*c*) clearly depends on the initial choice of the fixed point x_0. If another fixed point x_0 is chosen, then another isometry F of X into $\mathfrak{C}(X,R)$ is determined. It would seem, therefore, that there is little justification for calling the particular X^* defined in this problem *the* completion of X. In practice, however, we usually pursue the reasonable course of regarding isometric spaces as essentially identical. From this point of view, the X^* defined here is a complete metric space which contains X as a dense subspace; and since by (*h*) it is the only complete metric space with this property, it is natural to call it *the* completion of X.

[2] From this point on, we omit the adjective "ordered." It is to be understood that an n-tuple is *always* ordered.

We begin by defining addition and scalar multiplication in R^n. If $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$, then we define $x + y$ and αx (where α is any real number) by

$$x + y = (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)$$

and

$$\alpha x = (\alpha x_1, \alpha x_2, \ldots, \alpha x_n).$$

With the algebraic operations defined *coordinatewise* in this way, R^n is a real linear space. The origin or zero element is clearly $0 = (0, 0, \ldots, 0)$, and the negative of an element $x = (x_1, x_2, \ldots, x_n)$ is

$$-x = (-x_1, -x_2, \ldots, -x_n).$$

When we speak of R^n as an *n-dimensional* space, all we mean at this stage is that each element $x = (x_1, x_2, \ldots, x_n)$ is the ordered array of its n *coordinates* $x_1, x_2, \ldots, x_n$.

The reader is probably familiar with vector algebra in the ordinary three-dimensional space of our physical intuition. If so, then he is

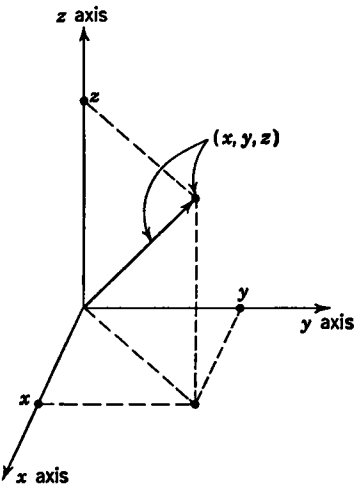


Fig. 21. A vector (or point) in ordinary space.

accustomed to regarding a point in this space as being essentially identical with the arrow (or vector) from the origin to that point, in the sense that given the point, the vector is determined, and given the vector, the point is determined. This situation is illustrated in Fig. 21. The above definitions of addition and scalar multiplication in R^n correspond to vector addition and the multiplication of a vector by a real number. A word of warning must be given. In ordinary vector algebra, a vector is usually allowed to have its tail at any point in the space and its head at any

other point. It should be clearly understood, however, that for us a vector always has its tail at the origin. In accordance with this intuitive picture, we may think of the elements of the real linear space R^n either as points or as generalized vectors from the origin to those points. The latter view is often more fruitful and illuminating.

There is yet a third interpretation of the elements of R^n, of great significance from the point of view of generalizations. An n-tuple $x = (x_1, x_2, \ldots, x_n)$ of real numbers can be thought of as a real function f defined on the set $\{1, 2, \ldots, n\}$ of the first n positive integers. The ith coordinate x_i of x is then just the value of this function at the integer i ($f(i) = x_i$), and the coordinatewise operations defined above become pointwise operations. This way of thinking about the elements of R^n should help to allay any doubts which might be felt as to the feasibility of visualizing n-dimensional spaces for $n \geq 4$. The four-dimensional space R^4, for instance, is merely the space of all real functions defined on the set consisting of the first four positive integers, and there is surely nothing mysterious or incomprehensible about this. The advantages of the function notation are so great that we shall often (but not always) use it in preference to the n-tuple notation. The reader will find it profitable to keep in mind all three aspects of the elements of R^n—as points, as vectors, and as functions—and he will train himself to use that interpretation (and notation) which appears most natural in any given situation.

Our next task is to define a suitable norm on the linear space R^n. We recall that in solid analytic geometry the usual distance from a point (x, y, z) to the origin (see Fig. 21) is given by the expression

$$\sqrt{x^2 + y^2 + z^2}.$$

If $x = (x_1, x_2, \ldots, x_n)$ is an arbitrary element of R^n, then it is natural to define $\|x\|$—the distance from the point x to the origin, or the length of the vector x—by

$$\begin{aligned} \|x\| &= \sqrt{|x_1|^2 + |x_2|^2 + \cdots + |x_n|^2} \\ &= \Big(\sum_{i=1}^{n} |x_i|^2\Big)^{1/2}. \end{aligned}$$

If we think of R^n as composed of real functions f defined on $\{1, 2, \ldots, n\}$, then this definition becomes

$$\|f\| = \Big(\sum_{i=1}^{n} |f(i)|^2\Big)^{1/2}.$$

This is called the *Euclidean norm* on R^n, and the real linear space R^n normed in this way is called *n-dimensional Euclidean space.* The

Euclidean plane is the real linear space R^2 with its Euclidean norm; that is, it is the coordinate plane equipped with the above algebraic operations and the above norm. For reasons which will appear a little later, we observe that our formula defining $\|x\|$ can be applied equally well to n-tuples of complex numbers.

We have not yet proved, of course, that the above expression for $\|x\|$ possesses the three properties required by the definition of a norm. The first and third of these conditions are clearly satisfied. The second, namely, that

$$\|x + y\| \leq \|x\| + \|y\|,$$

is another matter. We prove this by the following two lemmas, of which the first is essentially a tool used in the proof of the second.

Lemma (Cauchy's Inequality). *Let* $x = (x_1, x_2, \ldots, x_n)$ *and* $y = (y_1, y_2, \ldots, y_n)$ *be two n-tuples of real or complex numbers. Then*

$$\sum_{i=1}^{n} |x_i y_i| \leq \Big(\sum_{i=1}^{n} |x_i|^2\Big)^{1/2} \Big(\sum_{i=1}^{n} |y_i|^2\Big)^{1/2},$$

or, in our notation, $\Sigma_{i=1}^{n} |x_i y_i| \leq \|x\| \, \|y\|$.

PROOF. We first remark that if a and b are any two non-negative real numbers, then $a^{1/2} b^{1/2} \leq (a + b)/2$; for on squaring both sides and rearranging, this is equivalent to $0 \leq (a - b)^2$, which is obviously true. If $x = 0$ or $y = 0$, the assertion of the lemma is clear. We therefore assume that $x \neq 0$ and $y \neq 0$. We define a_i and b_i by $a_i = (|x_i|/\|x\|)^2$ and $b_i = (|y_i|/\|y\|)^2$. By the above remark, we obtain the following for each i:

$$\frac{|x_i y_i|}{\|x\| \, \|y\|} \leq \frac{|x_i|^2/\|x\|^2 + |y_i|^2/\|y\|^2}{2}.$$

Summing these inequalities as i varies from 1 to n yields

$$\frac{\sum_{i=1}^{n} |x_i y_i|}{\|x\| \, \|y\|} \leq \frac{1 + 1}{2} = 1,$$

from which our conclusion follows at once.

Lemma (Minkowski's Inequality). *Let* $x = (x_1, x_2, \ldots, x_n)$ *and* $y = (y_1, y_2, \ldots, y_n)$ *be two n-tuples of real or complex numbers. Then*

$$\Big(\sum_{i=1}^{n} |x_i + y_i|^2\Big)^{1/2} \leq \Big(\sum_{i=1}^{n} |x_i|^2\Big)^{1/2} + \Big(\sum_{i=1}^{n} |y_i|^2\Big)^{1/2},$$

or, in our notation, $\|x + y\| \leq \|x\| + \|y\|$.

PROOF. Using Cauchy's inequality, we have the following chain of relations:

$$\begin{aligned}\|x+y\|^2 &= \sum_{i=1}^{n} |x_i + y_i|\,|x_i + y_i| \\ &\leq \sum_{i=1}^{n} |x_i + y_i|(|x_i| + |y_i|) \\ &= \sum_{i=1}^{n} |x_i + y_i|\,|x_i| + \sum_{i=1}^{n} |x_i + y_i|\,|y_i| \\ &\leq \|x + y\|\,\|x\| + \|x + y\|\,\|y\| \\ &= \|x + y\|(\|x\| + \|y\|),\end{aligned}$$

or summarizing,

$$\|x + y\|^2 \leq \|x + y\|(\|x\| + \|y\|).$$

If $\|x + y\| = 0$, our lemma is trivially true; otherwise, it follows from the inequality last written on dividing through by $\|x + y\|$.

We are now in a position to state

Theorem A. *The set R^n of all n-tuples $x = (x_1, x_2, \ldots, x_n)$ of real numbers is a real Banach space with respect to coordinatewise addition and scalar multiplication and the norm defined by $\|x\| = (\Sigma_{i=1}^{n} |x_i|^2)^{1/2}$.*

PROOF. In view of the above discussions, all that remains is to prove completeness. It will be convenient here to use the function notation, so that a typical element of our space is regarded as a real function defined on $\{1, 2, \ldots, n\}$. Let $\{f_m\}$ be a Cauchy sequence in R^n. If $\epsilon > 0$ is given, then for all sufficiently large m and m' we have $\|f_m - f_{m'}\| < \epsilon$, $\|f_m - f_{m'}\|^2 < \epsilon^2$, and $\Sigma_{i=1}^{n} |f_m(i) - f_{m'}(i)|^2 < \epsilon^2$; and from this we see that $|f_m(i) - f_{m'}(i)| < \epsilon$ for each i (and all sufficiently large m and m'). The sequence $\{f_m\}$ therefore converges pointwise to a limit function f defined by $f(i) = \lim f_m(i)$. Since the set $\{1, 2, \ldots, n\}$ is finite, this convergence is uniform. We can thus find a positive integer m_0 such that $|f_m(i) - f(i)| < \epsilon/n^{1/2}$ for all $m \geq m_0$ and every i. Squaring each of these inequalities and summing as i varies from 1 to n yields $\Sigma_{i=1}^{n} |f_m(i) - f(i)|^2 < \epsilon^2$ or $\|f_m - f\| < \epsilon$ for all $m \geq m_0$. This shows that the Cauchy sequence $\{f_m\}$ converges to the limit f, so R^n is complete.

Just as in the previous section, virtually every statement we have made about n-tuples of real numbers (or about real functions defined on $\{1, 2, \ldots, n\}$) has its complex analogue. We therefore consider the following theorem to be fully proved.

Theorem B. *The set C^n of all n-tuples $z = (z_1, z_2, \ldots, z_n)$ of complex numbers is a complex Banach space with respect to coordinatewise addition and scalar multiplication and the norm defined by $\|z\| = (\Sigma_{i=1}^{n} |z_i|^2)^{1/2}$.*

The space C^n, with these algebraic operations and this norm, is called *n-dimensional unitary space.* Needless to say, it can equally well be viewed as the set of all complex functions f defined on the set $\{1, 2, \ldots, n\}$, with addition and scalar multiplication (by complex scalars) defined pointwise and the norm defined by $\|f\| = (\Sigma_{i=1}^{n} |f(i)|^2)^{1/2}$.

The four spaces defined and discussed in this and the previous section—$\mathcal{C}(X,R)$ and $\mathcal{C}(X,C)$, and R^n and C^n—form the foundation for all our future work. In Chaps. 3 to 7 we generalize the first two by loosening the restrictions on the underlying space X. In Chaps. 9 to 11 we study all four from a wider point of view, with special emphasis on C^n. And in the last three chapters we pull these lines of development together in such a way that each aspect of our work sheds light on all the others.

Problems

1. Show that a non-empty subset A of R^n is bounded $\Leftrightarrow$ there exists a real number K such that for each $x = (x_1, x_2, \ldots, x_n)$ in A we have $|x_i| \leq K$ for every subscript i.
2. Let X be the set $\{1, 2, \ldots, n\}$, equipped with the metric defined in Example 9-1. Then $\mathcal{C}(X,R)$ and R^n are two Banach spaces which are essentially identical as real linear spaces but which have different norms. Show that they have the same open sets.
3. Prove the following extension of Minkowski's inequality. If $x = \{x_1, x_2, \ldots, x_n, \ldots\}$ and $y = \{y_1, y_2, \ldots, y_n, \ldots\}$ are two sequences of real or complex numbers such that $\Sigma_{n=1}^{\infty} |x_n|^2$ and $\Sigma_{n=1}^{\infty} |y_n|^2$ are convergent, then $\Sigma_{n=1}^{\infty} |x_n + y_n|^2$ is also convergent, and

$$\Big(\sum_{n=1}^{\infty} |x_n + y_n|^2\Big)^{1/2} \leq \Big(\sum_{n=1}^{\infty} |x_n|^2\Big)^{1/2} + \Big(\sum_{n=1}^{\infty} |y_n|^2\Big)^{1/2}.$$

   This statement is also called Minkowski's inequality—for infinite sums.
4. The set of all sequences $x = \{x_1, x_2, \ldots, x_n, \ldots\}$ of real numbers such that $\Sigma_{n=1}^{\infty} |x_n|^2$ converges is denoted by R^∞. If addition and scalar multiplication are defined coordinatewise (or termwise), and if a norm is defined by $\|x\| = (\Sigma_{n=1}^{\infty} |x_n|^2)^{1/2}$, show that R^∞ is a real Banach space. R^∞ is called *infinite-dimensional Euclidean space.* The *infinite-dimensional unitary space* C^∞ is defined similarly, and is a complex Banach space.